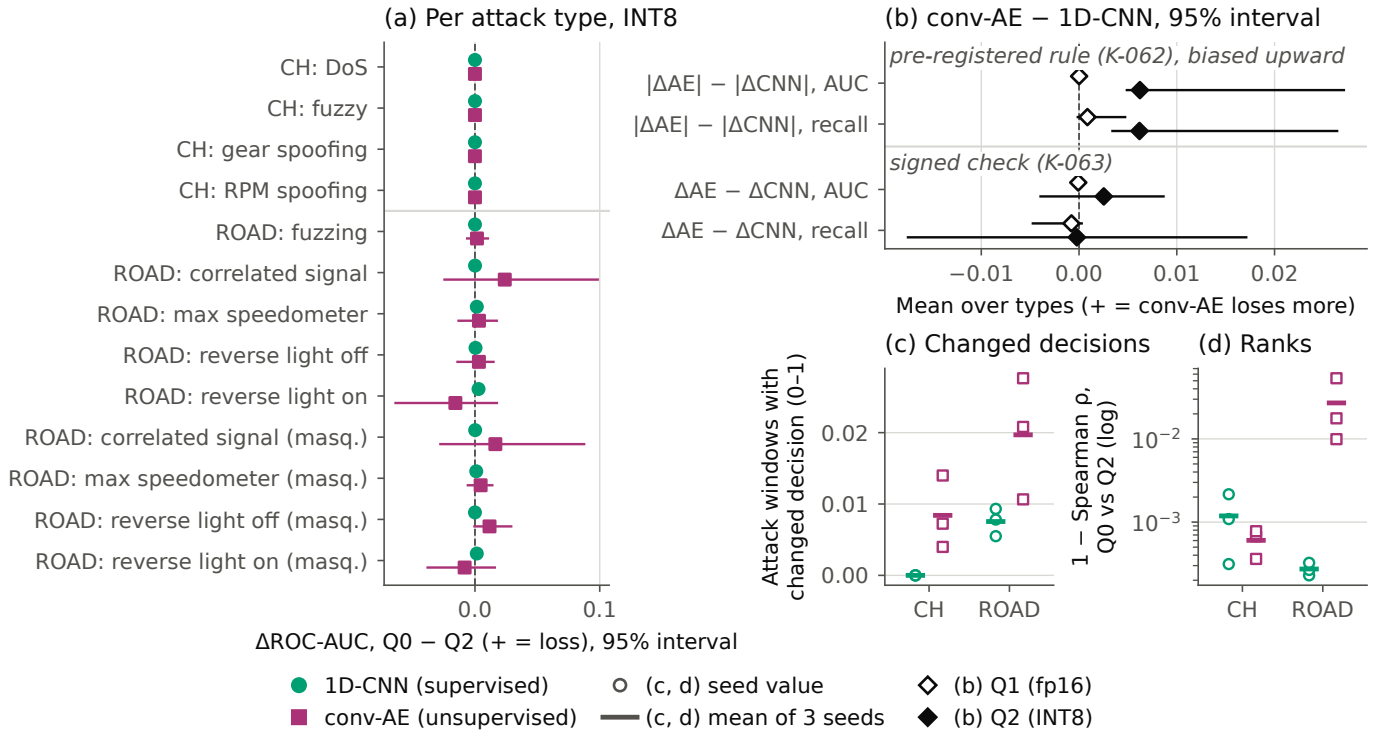
(a) Per attack type, INT8
CH: DoS
CH: fuzzy
CH: gear spoofing
CH: RPM spoofing
ROAD: fuzzing
ROAD: correlated signal
ROAD: max speedometer
ROAD: reverse light off
ROAD: reverse light on
ROAD: correlated signal (masq.)
ROAD: max speedometer (masq.)
ROAD: reverse light off (masq.)
ROAD: reverse light on (masq.)
0.0
0.1
ΔROC-AUC, Q0 − Q2 (+ = loss), 95% interval
(b) conv-AE − 1D-CNN, 95% interval
pre-registered rule (K-062), biased upward
|ΔAE| − |ΔCNN|, AUC
|ΔAE| − |ΔCNN|, recall
signed check (K-063)
ΔAE − ΔCNN, AUC
ΔAE − ΔCNN, recall
−0.01
0.00
0.01
0.02
Mean over types (+ = conv-AE loses more)
(c) Changed decisions
Attack windows with changed decision (0–1)
0.00
0.01
0.02
CH
ROAD
(d) Ranks
1 − Spearman ρ, Q0 vs Q2 (log)
10^{-2}
10^{-3}
CH
ROAD
1D-CNN (supervised)
conv-AE (unsupervised)
(c, d) seed value
(c, d) mean of 3 seeds
(b) Q1 (fp16)
(b) Q2 (INT8)